

120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 9: Supervised Learning

Day 61: Feature Engineering from Market Data

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1 Introduction

In quantitative finance, raw market data rarely contains direct predictive signals. Instead, meaningful patterns must be extracted through a process known as **feature engineering**. Feature engineering transforms raw financial data into structured variables that can be used as inputs for machine learning models.

In a supervised learning framework, we define a feature matrix:

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,n} \\ x_{2,1} & x_{2,2} & \dots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{T,1} & x_{T,2} & \dots & x_{T,n} \end{bmatrix}$$

where each row represents observations at time t , and each column represents a feature. The goal is to use these features to predict a target variable such as future returns:

$$y_t = r_{t+1}$$

where r_t represents the asset return.

Feature engineering is one of the most important stages in financial modeling because well-designed features capture economic intuition, market behavior, and statistical structure within financial time series.

2 Return-Based Features

Returns are the fundamental variables used in financial modeling. Instead of modeling prices directly, most quantitative models focus on returns because they tend to be more stationary and statistically manageable.

2.1 Log Returns

Logarithmic returns are widely used in finance due to their desirable mathematical properties.

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

where:

- P_t is the price at time t
- P_{t-1} is the previous price

Log returns are approximately additive across time:

$$r_{t,t+k} \approx r_t + r_{t+1} + \cdots + r_{t+k}$$

This property simplifies many financial models.

2.2 Rolling Returns

Momentum signals often rely on cumulative returns over multiple periods.

$$r_{t,k} = \ln \left(\frac{P_t}{P_{t-k}} \right)$$

Common horizons include:

- 5-day returns
- 20-day returns
- 60-day returns

These capture short-term and medium-term trends in asset prices.

3 Trend and Momentum Features

Momentum is one of the strongest empirical anomalies documented in financial markets.

3.1 Moving Averages

A simple moving average (SMA) smooths price data over a window of length k .

$$MA_t = \frac{1}{k} \sum_{i=0}^{k-1} P_{t-i}$$

Moving averages help identify trends by filtering out short-term noise.

3.2 Price Relative to Moving Average

A common signal compares the current price with its moving average:

$$Signal_t = \frac{P_t}{MA_t} - 1$$

If:

$$Signal_t > 0$$

the asset is trading above its trend.

If:

$$Signal_t < 0$$

the asset is trading below its trend.

This indicator is widely used in trend-following strategies.

4 Volatility Features

Volatility is a key component of financial markets. Unlike returns, volatility tends to exhibit strong persistence and clustering.

4.1 Rolling Volatility

Volatility can be estimated using a rolling standard deviation:

$$\sigma_t = \sqrt{\frac{1}{k} \sum_{i=1}^k (r_{t-i} - \bar{r})^2}$$

where:

- k is the rolling window size
- $\bar{r}$ is the mean return in the window

High volatility often corresponds to market stress or uncertainty.

5 Volume-Based Features

Trading volume provides additional information about market participation and liquidity.

A simple volume-based signal is the percentage change in trading volume:

$$\Delta V_t = \frac{V_t - V_{t-1}}{V_{t-1}}$$

Large volume spikes often confirm price movements or indicate strong market interest.

6 Technical Indicators

Many feature engineering techniques originate from technical analysis.

Examples include:

6.1 Relative Strength Index (RSI)

The RSI measures the speed and magnitude of price changes.

$$RSI = 100 - \frac{100}{1 + RS}$$

where RS is the ratio of average gains to average losses.

6.2 Bollinger Bands

Bollinger Bands use volatility to construct dynamic price boundaries.

$$Upper = MA + 2\sigma$$

$$Lower = MA - 2\sigma$$

Prices approaching these bands may indicate overbought or oversold conditions.

7 Cross-Asset Features

Financial markets are interconnected. Information from other assets often improves predictive models.

Examples include:

- Yield curve slope
- Credit spreads
- Volatility index (VIX)
- Currency strength
- Commodity prices

Cross-asset signals are commonly used by macro hedge funds and systematic trading strategies.

8 Avoiding Look-Ahead Bias

One of the most critical issues in financial modeling is avoiding **look-ahead bias**. A model must only use information available at the time of prediction.

Formally:

$$X_t \subseteq \{P_{t-1}, P_{t-2}, \dots\}$$

The target variable is typically:

$$y_t = r_{t+1}$$

Using future data in feature construction would artificially inflate model performance.

9 Feature Scaling

Machine learning algorithms often require normalized features.

A common scaling method is standardization:

$$x'_i = \frac{x_i - \mu}{\sigma}$$

where:

- μ is the mean of the feature
- σ is the standard deviation

This transformation ensures features have comparable magnitudes.

10 Financial Interpretation

Good financial features should satisfy several principles:

- Capture economic intuition
- Reflect market structure
- Avoid information leakage
- Be statistically stable

In many real-world trading systems, the success of a model depends more on the quality of features than on the complexity of the algorithm itself.

11 Conclusion

Feature engineering is a central component of machine learning in finance. By transforming raw market data into meaningful predictors, researchers can extract signals related to momentum, volatility, liquidity, and cross-market relationships.

These engineered features form the foundation of predictive models used in algorithmic trading, risk management, and portfolio construction.